

CHEMICAL STRUCTURES, REPRESENTATIONS AND DRAWING TOOLS

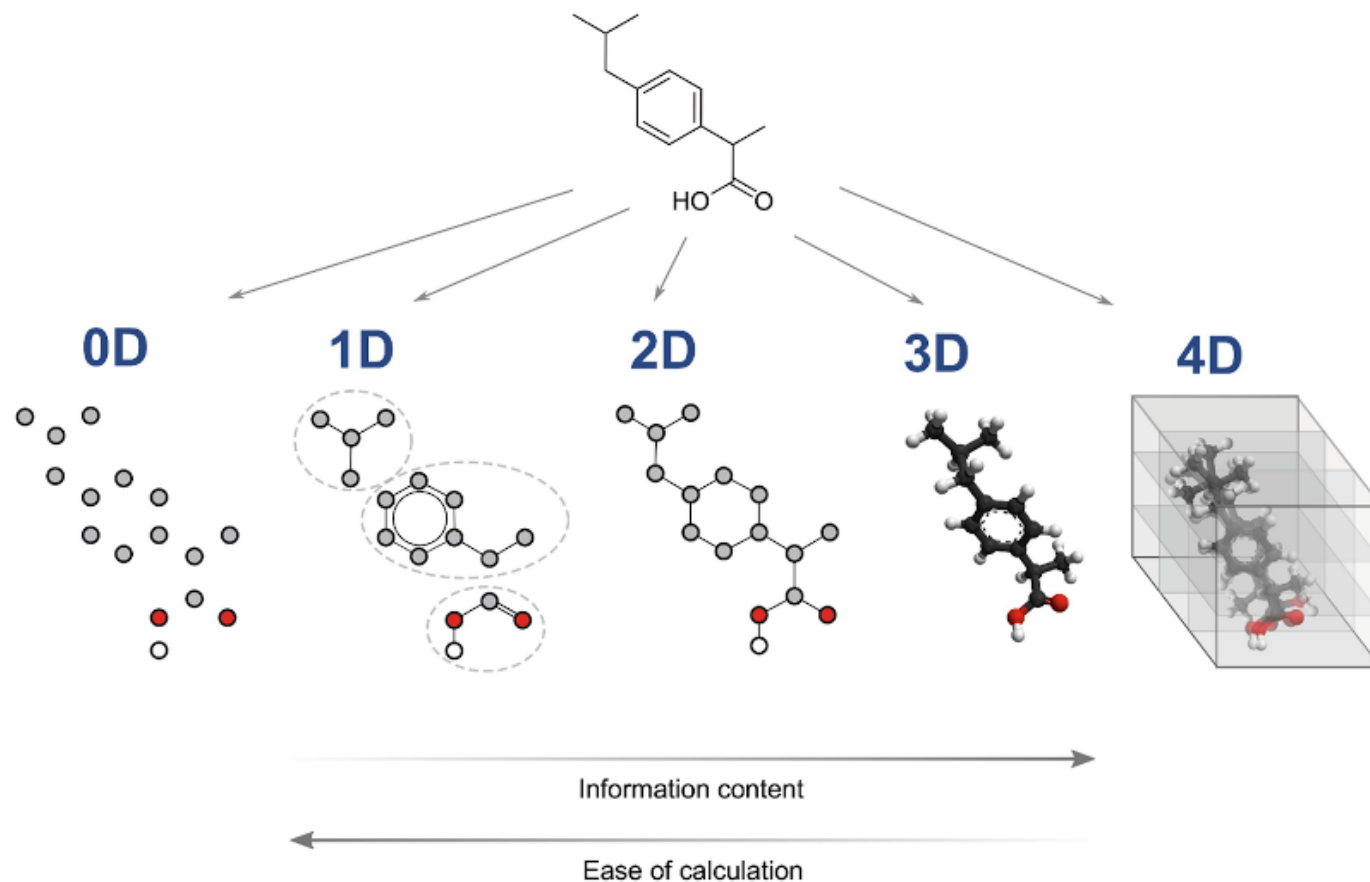
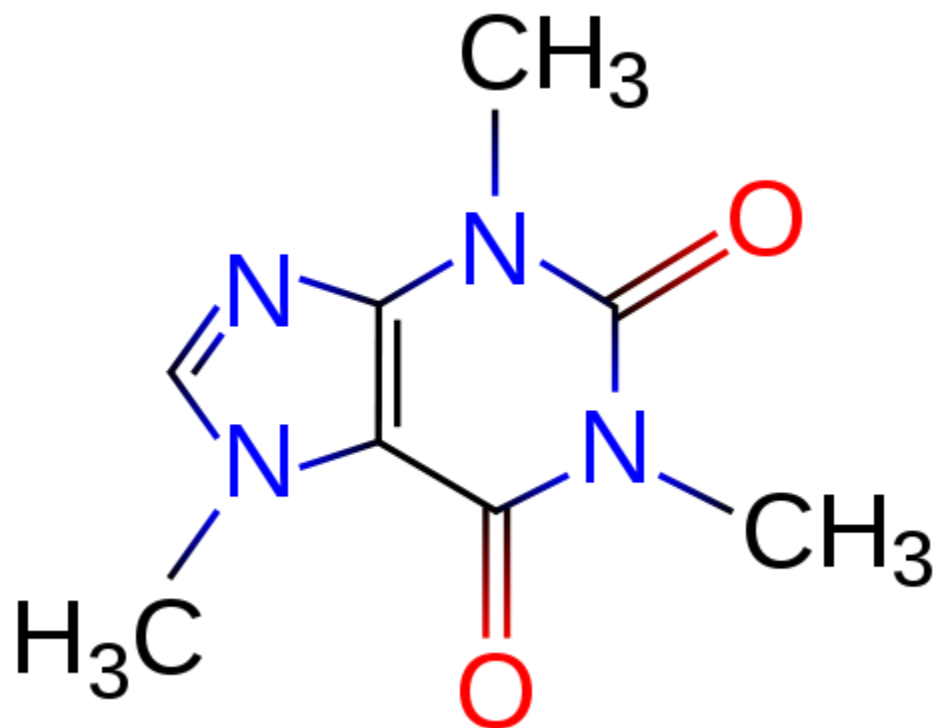
Dr. Oluyemi Wande | Dr. Adeniyi Adewumi | Shadrach Eze

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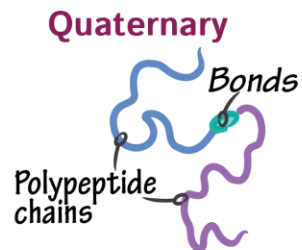
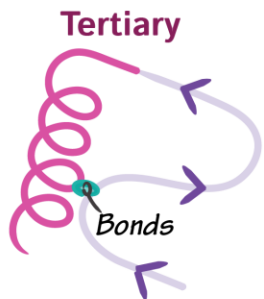
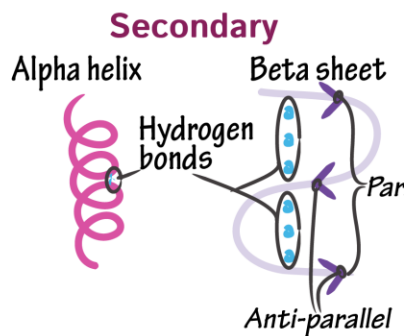
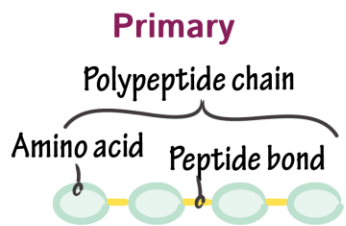
CONTENTS

- Chemical structure
- Protein Structure
- Chemical/Molecular Representations
- File formats and representations (pdb mol mol2 sdf pdbqt etc.); Meaning, Uses and applications
- Use of chemsketch, chemdraw and other online serve tools for drawing and chemical enumeration
- Convert between file types; Open Babel

Chemical Structure



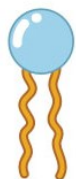
CLASSES OF PROTEIN STRUCTURE



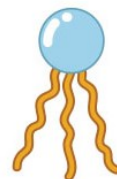
Cleveland
Clinic
©2022

Lipids

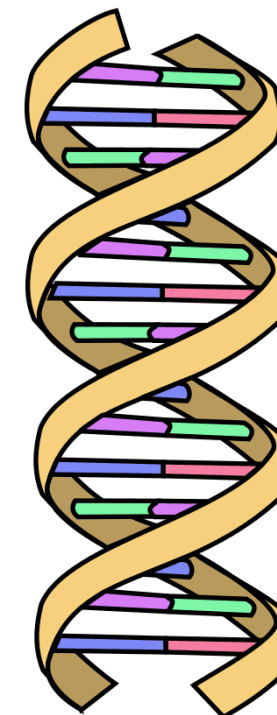
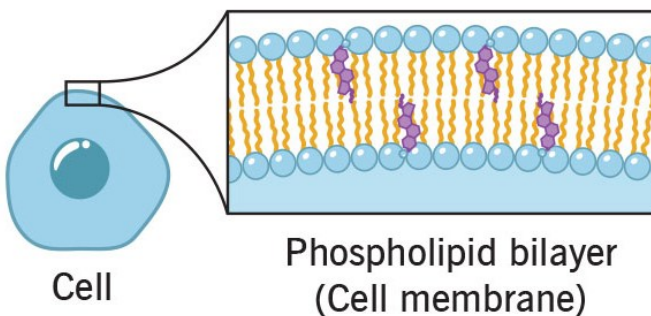
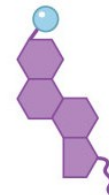
Phospholipid



Triglyceride



Cholesterol



DNA

— = Adenine

— = Thymine

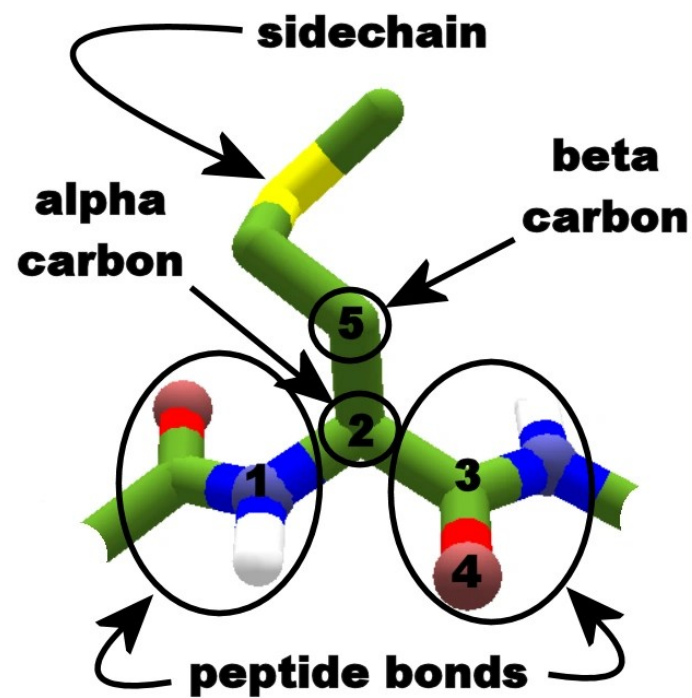
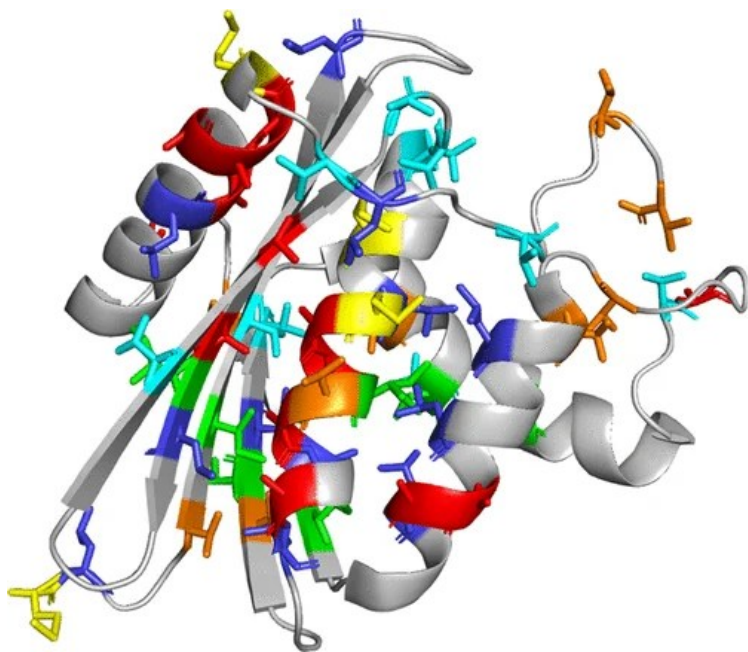
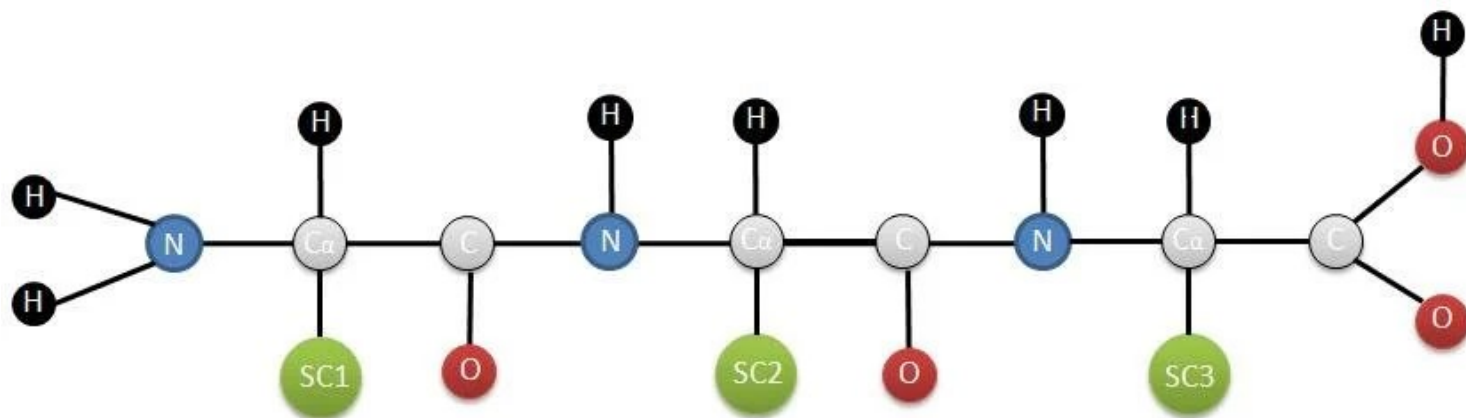
— = Cytosine

— = Guanine

— = Phosphate
backbone

PROTEIN AND MACROMOLECULAR STRUCTURE

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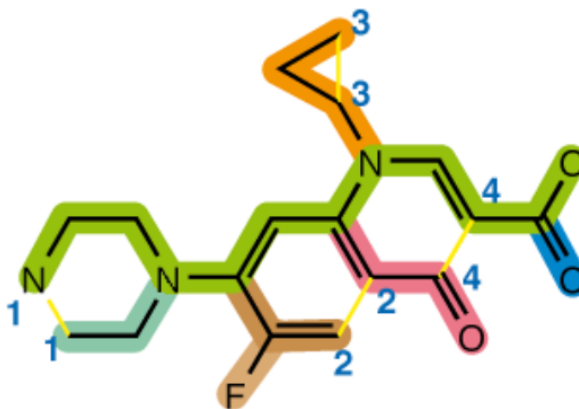
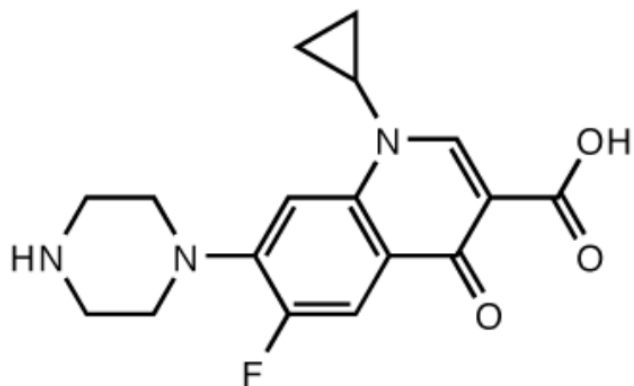
Chemical/Molecular Representations

Generic names ⁵	Dicycloverine HCl, benacol, bentyl, dibent, Dyspas, and so on
Mol. formula	C₁₉H₃₆ClNO₂
IUPAC name	2-(Diethylamino)ethyl 1-cyclohexylcyclohexane-1-carboxylate hydrochloride
CAS RN	67 – 92 – 5
Canonical SMILES	<chem>CCN(CC)CCOC(=O)C1(CCCCC1)C2CCCCC2.Cl</chem>
InChI	InChI = 1S/C19H35NO2.ClH/c1-3-20(4-2)15-16-22-18(21)19 (13-9-6-10-14-19)17-11-7-5-8-12-17;/h17H,3-16H2,1-2H3;1H InChIKey:GUBNMFJOJGDCEL-UHFFFAOYSA-N
WLN ⁶	L6TJA-AL6TJAVO2N2&2&GH

Chemical/Molecular Representations

Simplified Molecular-Input Line-Entry System

Smiles Notation



N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O



Why Smiles Notation

Very
Flexible

Computer
readable

Universally
Accepted

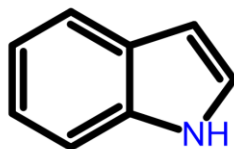
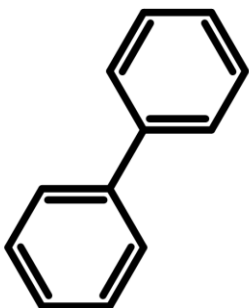
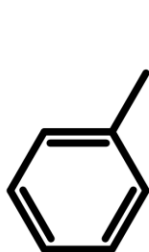
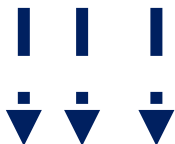
Indepth
information

The SMARTS Pattern

SMILES ARbitrary Target Specification (SMARTS)

Allows one to query a database of molecules that match a certain pattern

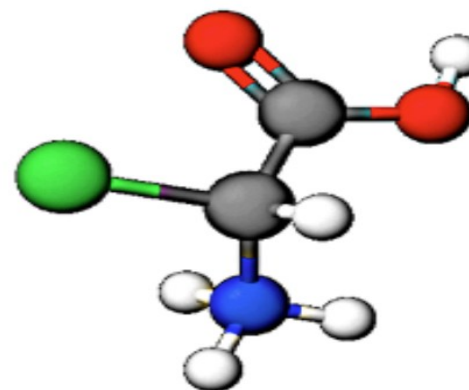
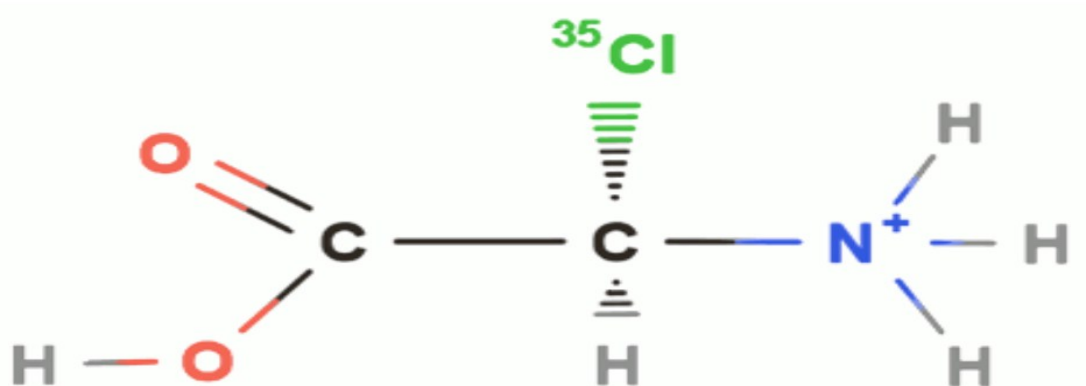
```
from rdkit import Chem  
mol = Chem.MolFromSmarts("c1ccccc1[#6]")
```



Primitive	Property	Notes
"	"single or aromatic"	"unspecified bonds"
-	single	
=	double	
#	triple	
:	aromatic	
~	"any bond"	
@	"ring bond"	
/	"directional"	
\	"directional"	
->	"dative right"	extension
<-	"dative left"	extension

Primitive	Property	"Default value"	Range?	Notes
a	"aromatic atom"			
A	"aliphatic atom"			
d	"non-hydrogen degree"	1	Y	extension
D	"explicit degree"	1	Y	
h	"number of implicit hs"	>0	Y	
H	"total number of Hs"	1		
r	"size of smallest SSSR ring"	>0	Y	
R	"number of SSSR rings"	>0	Y	
v	"total valence"	1	Y	
x	"number of ring bonds"	>0	Y	
X	"total degree"	1	Y	

Chemical/Molecular Representations



The International
Chemical Identifier

InChI=1S/C2H4ClNO2/c3-1(4)2(5)6/h1H,4H2,(H,5,6)/p+1/t1-/m0/s1/i3+0

InChI version

Chemical formula sub-layer

Hydrogen atoms sub-layer

Stereochemical layer

InChIKey=UWPWWENWLZPQGU-WRFRXMDISA-O

Main layer

Stereochemistry & isotopes

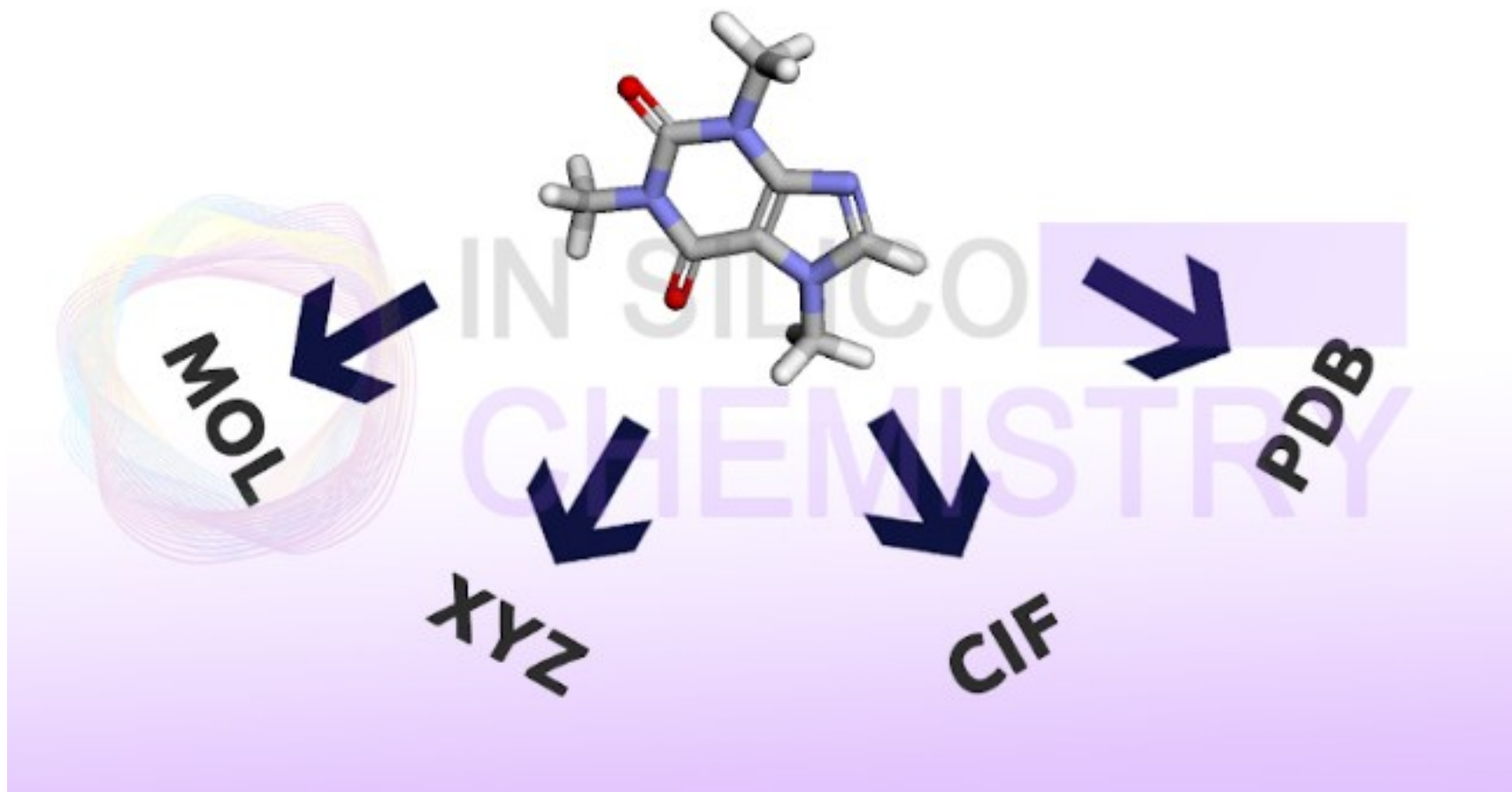
version

protons

InChI Notation

File Formats

- These are **specific implementations** that store chemical/molecular representations in files.
- It defines *how the **data is written** and **exchanged*** between programs.



File Formats

PDB File

Atomic Coordinates: PDB Format

	Amino Acid		Chain name		Sequence Number	-----Coordinates-----			
	Element					X	Y	Z	(etc.)
ATOM	1	N	ASP	L	1	4.060	7.307	5.186	...
ATOM	2	CA	ASP	L	1	4.042	7.776	6.553	...
ATOM	3	C	ASP	L	1	2.668	8.426	6.644	...
ATOM	4	O	ASP	L	1	1.987	8.438	5.606	...
ATOM	5	CB	ASP	L	1	5.090	8.827	6.797	...
ATOM	6	CG	ASP	L	1	6.338	8.761	5.929	...
ATOM	7	OD1	ASP	L	1	6.576	9.758	5.241	...
ATOM	8	OD2	ASP	L	1	7.065	7.759	5.948	...

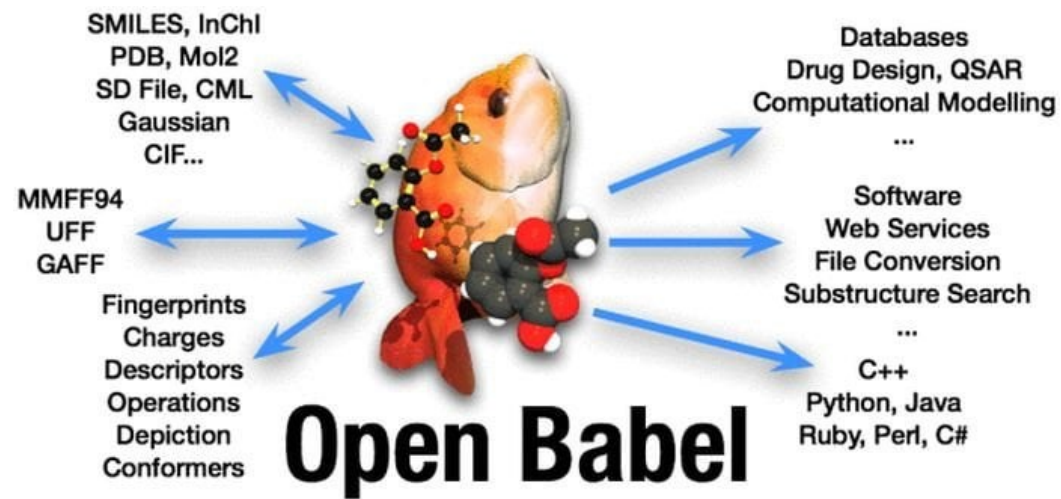
\\
Element position within amino acid

PDBQT File

REMARK						x	y	z	vdW	Elec	q	Type
REMARK												
ROOT												
ATOM	5	C26	***	A	401	19.818	-26.103	32.156	0.00	0.00	+0.000	A
ATOM	6	C02	***	A	401	21.136	-27.030	35.720	0.00	0.00	+0.000	C
ATOM	7	C03	***	A	401	22.390	-26.626	33.483	0.00	0.00	+0.000	A
ATOM	8	C04	***	A	401	22.376	-26.303	32.074	0.00	0.00	+0.000	A
ATOM	9	C05	***	A	401	23.628	-26.253	31.381	0.00	0.00	+0.000	C
ATOM	10	C06	***	A	401	24.852	-26.494	31.976	0.00	0.00	+0.000	C
ATOM	11	C07	***	A	401	24.865	-26.808	33.339	0.00	0.00	+0.000	C
ATOM	12	C08	***	A	401	23.688	-26.872	34.064	0.00	0.00	+0.000	C
ATOM	13	C09	***	A	401	21.014	-26.059	31.520	0.00	0.00	+0.000	A
ATOM	14	C11	***	A	401	18.972	-25.528	29.970	0.00	0.00	+0.000	C
ATOM	21	C25	***	A	401	18.672	-25.799	31.259	0.00	0.00	+0.000	C
ATOM	22	C27	***	A	401	19.817	-26.442	33.637	0.00	0.00	+0.000	A
ATOM	23	N01	***	A	401	21.162	-26.691	34.228	0.00	0.00	+0.000	N
ATOM	27	O28	***	A	401	18.852	-26.518	34.341	0.00	0.00	+0.000	OA
ATOM	28	S10	***	A	401	20.724	-25.646	29.837	0.00	0.00	+0.000	S
ENDROOT												
BRANCH	14	15										
ATOM	15	C12	***	A	401	18.154	-25.163	28.688	0.00	0.00	+0.000	C
ATOM	26	O24	***	A	401	18.712	-24.746	27.727	0.00	0.00	+0.000	OA
BRANCH	15	24										
ATOM	24	N13	***	A	401	16.722	-25.306	28.614	0.00	0.00	+0.000	NA
BRANCH	24	16										
ATOM	16	C14	***	A	401	15.799	-25.024	27.520	0.00	0.00	+0.000	C



Open-Source Cheminformatics
and Machine Learning



FILE CONVERSION

Installation



```
Sudo apt update  
Sudo apt install openbabel
```

```
conda install openbabel -c conda-forge
```

Help

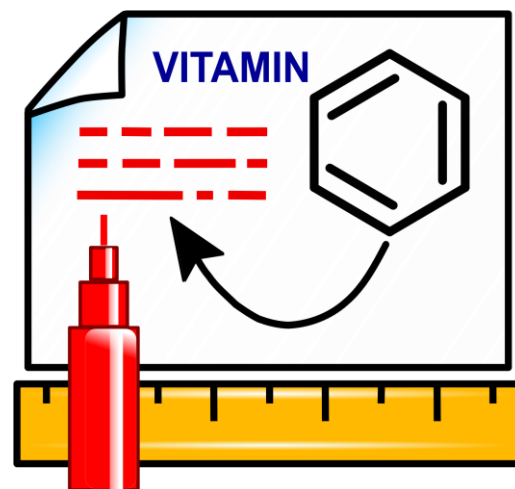


```
Obabel -H
```

Usage



```
Obabel -ipdb ligand.pdb -omol2 -O ligand.mol2
```



DRAWING CHEMICAL STRUCTURES

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EXTRA RESOURCES

- <https://scispace.com/papers/chemical-data-formats-fingerprints-and-other-molecular-3tk90zk5dl>
- https://www.scotchem.ac.uk/pldc/file_formats/#converting-between-file-types
- https://www.daylight.com/dayhtml_tutorials/languages/smarts/index.html
- O'Boyle, N.M., Banck, M., James, C.A. et al. Open Babel: An open chemical toolbox. J Cheminform 3, 33 (2011). <https://doi.org/10.1186/1758-2946-3-33>
- <https://practicalcheminformatics.blogspot.com/>